

132 Calibrated MTB Suspensions

Dual-chamber polytropic model · BikeLab SAG

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Abstract

132 MTB forks and shocks calibrated by brand, model and segment through inverse calibration against official pressure-by-weight tables. Basis of the dual-chamber polytropic model of the BikeLab SAG Calculator. Per-segment confidence via Monte Carlo (10,000 iterations).

Dataset composition (132 models)

Brand	Models	With ground truth	Segments
RockShox (SRAM)	45	12	Entry-DH + Legacy
Fox Racing Shox	29	10	Mid-DH + Legacy
SR Suntour	12	6	Entry-E-MTB
Manitou	8	0	Mid-DH
Öhlins	6	0	High (3 cámara)
Formula	6	0	Mid-High-High
Cane Creek	5	0	High
Marzocchi	4	0	Mid-Mid-High
DVO	4	0	Mid-High-Enduro
X-Fusion, MRP, RST, +	13	0	Entry-High

Sample — per-segment confidence

Segment	Representative models	MC confidence	Error [%SAG]
Entry (Solo Air)	RS Judy, SR Epixon	72-76%	1,3-1,5%
Mid (DebonAir/EQ)	Fox 34 Rhythm, RS 35	83%	1,1%
Mid-High (EVOL)	RS Pike Select, Fox 36	83-86%	1,0%
Shocks (sin GT)	RS Super Deluxe	42%	—

Dataset variables

brand; model; segment; official pressure by weight (PSI); psi/N ratio; leverage ratio; mean %SAG error; Monte Carlo confidence (%). The per-model calibrated curve and the inverse-calibration code are shared under agreement.

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Provisional by design. A verifiable sample is published; the full raw set and the reproducible method are shared under agreement. Contact: carlos.ravello@zoovettravel.com